

Relevance of Filter Bank Common Spatial Patterns Using Multiple Kernel Learning in Motor Imagery

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Abstract. Brain-Computer Interfaces directly communicate the human brain and machines through the analysis of sensorimotor activity, relying on the Motor Imagery paradigm of cognitive neuroscience. Conventional BCI systems use electroencephalographic signals due to its high temporal resolution, portability, and easiness to implement, for which the filter-banked analysis works as the characterization baseline. Due to such analysis yields to highly dimensional representation spaces leading to overtrained systems, we propose to combine the multiple spectral bands into a single representation space through the maximization of the centered kernel alignment criterion. As a result, the similarity between the measured EEG data and the available label sets is maximized, with the additional benefit of enhancing the spectral interpretation of the subject performance. The proposed κ-FB is evaluated in the dataset IIa of the BCI competition IV for a binary classification task. Attained accuracy proves that κ-FB outperforms other filter-banked representations without compromising the system confidence.

Keywords: Brain computer interfaces · Common spatial patterns Multiple kernel learning

1 Introduction

Brain-Computer Interfaces (BCI) take advantage of the extracted information from brain activity signals to set direct communication between the human brain and machines. BCI is used to help people with disability through the analysis of human sensorimotor functions, relying on the Motor Imagery (MI) paradigm of cognitive neuroscience. Among the non-invasive techniques for measuring brain activity, like magnetoencephalography (MEG), electroencephalography (EEG) is preferred due to its portability, easiness to implement, and higher temporal resolution.

Nonetheless, BCI systems are heavily dependent on useful feature extraction from EEG data, being Common spatial patterns (CSP) the most popular technique for MI classification tasks. CSP technique constructs a set of spatial filters

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that maximize the variance in a specific class, minimizing at the same time the variance of another class. Generally, classification accuracy depends on each subject, which may have different spectral behavior. To overcome this issue, Filter Bank Common Spatial Patterns (FB) approach is employed, however, at the cost of providing larger feature sets, which may lead to the curse of dimensionality effect.

Aiming to solve the issue as mentioned above, some spectral band selection algorithms can be applied: discriminant FBCSP (DFBCSP) using Fisher ratio [9], sliding window discriminative CSP (SWDCSP) by exploiting affinity propagation [8], sparse regression to select the significant CSP features according to the provided labels [12] or its extended approach sparse Bayesian learning (SBLFB) scheme [11]. Nevertheless, most of these feature selection methods are still computationally expensive. Moreover, in most of the cases, there is not a suitable validation framework that allows ensuring a generalized performance, leading to overtrained systems [10].

Here, we introduce a CSP spectral band selection method that computes the discriminative relevance of spectral band features, reflecting their contribution to discriminate the considered motor imagery classes. Instead of selecting one specific kernel function to encode the whole spectral band information, the proposed Multiple kernel learning (κ-FB) combines several single kernels (i.e., one kernel per spectral band) through a weighted sum, attempting to maximize the similarity between the measured EEG data and the available set of labels (i.e., prior knowledge). Using a kernel alignment algorithm to match the training spaces, the resulting relevance weights are interpreted as the contribution of each spectral band for improving the performance of the motor imagery task classification.

2 Materials and Methods

2.1 Feature Extraction Using Common Spatial Patterns

In binary classification tasks, CSP finds a spatial filter matrix $\boldsymbol{W}\in\mathbb{R}^{C\times 2K}$ to linearly map the EEG channel signals $\boldsymbol{X}\in\mathbb{R}^{C\times T}$ onto a space $\widetilde{\boldsymbol{X}} = \boldsymbol{W}^\top\boldsymbol{X}$, so that variance of the mapped signal is maximized for one class while the variance of another class is minimized. The spatial filters $\boldsymbol{w}^*\in\mathbb{R}^C$ to extract MI features are the solution of maximizing the Rayleigh quotient:

$$\boldsymbol{w}^* = \max_{\boldsymbol{w}} \frac{\boldsymbol{w}^\top\boldsymbol{\Sigma}^-\boldsymbol{w}}{\boldsymbol{w}^\top\boldsymbol{\Sigma}^+\boldsymbol{w}}, \text{ s.t.: } \|\boldsymbol{w}\|_2 = 1 \tag{1}$$

the spatial covariance matrix of the class $l\in\{-,+\}$ is estimated as $\boldsymbol{\Sigma}^l = \mathbb{E}\left\{\boldsymbol{X}_n\boldsymbol{X}_n^\top : n\in N_l\right\}$, being N_l the number of trials in class l. Notations $\|\cdot\|_2$ and $\mathbb{E}\{\cdot\}$ stand for ℓ_2-norm and expectation operator, respectively.

The optimization framework in Eq. (1) can be equivalently transformed into the generalized eigenvalue problem $\boldsymbol{\Sigma}^-\boldsymbol{w}^* = \lambda\boldsymbol{\Sigma}^+\boldsymbol{w}^*$ with $\lambda\in\mathbb{R}^+$. Thus, a set

of spatial filters $\boldsymbol{W}^* = [\boldsymbol{w}_1^* \dots \boldsymbol{w}_{2K}^*]$ can be obtained by collecting eigenvectors corresponding to the K largest and smallest eigenvalues of the generalized eigenvalue problem. To account further for brain activity at different spectral bandwidths, each n-trial of input MI data is bandpass filtered within a set of F frequency bands $\{\boldsymbol{X}_{n,f}{:}f{\in}F\}$. The CSP feature vector is then formed as $\boldsymbol{\xi} = [\xi_k{:}k{\in}2K]$ with $\boldsymbol{\xi}{\in}\mathbb{R}^{2FK}$, having entries $\xi_{n,k}^f = \ln(\mathrm{var}\{\boldsymbol{w}_k^{*\top}\boldsymbol{X}_{n,f}\})$, where $\mathrm{var}\{\cdot\}$ stands for the variance operator.

2.2 Multiple Kernel Learning for Filter-Bank Representation

To measure the pairwise proximity between trials at frequency band f, we perform the dot product-based kernel function as a measure of similarity between functionals, resulting in a set of kernels $\{\kappa(\boldsymbol{\xi}_n^f, \boldsymbol{\xi}_m^f){\in}\mathbb{R}^+{:}f{\in}F\}$ that must be properly combined to work out a unique similarity value for each pairwise trial comparison. We address this issue through a Multiple Kernel Learning (MKL) framework by the following weighted sum [5]:

$$\kappa(\boldsymbol{\xi}_n, \boldsymbol{\xi}_m; \boldsymbol{\mu}) = \sum_{f\in F} \mu_f \kappa(\boldsymbol{\xi}_n^f, \boldsymbol{\xi}_m^f), \tag{2a}$$

$$\text{s.t.:} \sum_{f\in F} \mu_f = 1,\ \mu_f \geq 0 \tag{2b}$$

where vector $\boldsymbol{\mu}{\in}\mathbb{R}^F$ holds the mixture weights that must be optimized to improve discrimination between classes across the label set. Under the constraints in Eq. (2b) to reproduce a convex function in Eq. (2a), optimization of $\boldsymbol{\mu}$ is carried out by the maximization of Centered Kernel Alignment, so that each CSP feature set is weighted according to its discrimination capability: the higher the weight, the more significant its contribution to the classifier performance.

As a result, the proposed approach using Multiple kernel learning for filter-bank representation in Eqs. (2a) and (2b) (noted as κ-FB) allows selecting the CSP feature sets extracted from the most relevant frequency bands for discriminating a motor imagery task at hand.

3 Experimental Set-Up

3.1 EEG Dataset and Preprocessing

The κ-FB approach is evaluated on a BCI competition IV dataset IIa[1], holding EEG data recorded by 22-electrode montage. Nine subjects were instructed to perform four MI tasks ("left hand","right hand", "both feet", and"tongue"). On different days, each subject completed two sessions, each one including six runs to perform 48 trials (i.e., 12 trials per class) and resulting in 288 trials per sitting. Here, we consider just the bi-class classification task: "left" versus"right hand".

[1] available at http://www.bbci.de/competition/iv/.

In the preprocessing stage, all EEG recordings are band-pass filtered between [0.5–100] Hz and sampled at 250 Hz, followed by a fifth order Butterworth band-pass filter to remove noises over 40 Hz and slow baseline signal under 4 Hz. Sub-band decomposition of each channel is further performed, for which the linearly distributed filters have been heuristically adjusted to 7 Hz bandwidth and 90% overlap. In total, we calculate 42 fifth-order Butterworth band-pass filters as to cover the whole bandwidth from4 to 40 Hz. To focus on the learning part of the MI task, each trial recording is temporarily segmented, extracting for analysis the interval that ranges from 0.5 s to 4.5 s. Lastly, CSP-based feature extraction is accomplished from the preprocessed recordings, yielding 42 feature vectors of dimension 22 per trial. However, we only take the first and the last three vectors [3].

3.2 Evaluation Scheme and Performance Assessment

Due to the provided universal approximating property, the κ-FB approach is implemented by a Gaussian kernel, for which the bandwidth parameter is computed as in [1]. For the sake of comparison, κ-FB is contrasted with a couple of similar approaches: The baseline Filter Bank Common Spatial Pattern (or FB) that aims to perform the selection of critical spatio-temporal discriminative characteristics from diverse frequency bands [2]; and its spare version (s-FB) that implements a sparse regression of filter-bank CSP features to predict the trial labels [12].

The classifier is evaluated on a kernelized k-nearest neighbors (Kk-NN) machine, for which the number of neighbors is tuned by an exhaustive search of the best average validation performance. As a measure of classifier performance, we estimate the average accuracy (a_c) by a five-fold cross-validation scheme.

4 Results and Discussion

By handling a convex combination of introduced functional kernels, here, the feature selection of Filter Bank Common Spatial Patterns is performed for improving the discrimination capability in Motor Imagery tasks. During validation of κ-FB approach, the following findings come out from the experiments:

To improve discrimination across MI classes, a multiple kernel learning framework is devised for optimizing the mixture weights to work out a unique similarity value between trials at each frequency band. Optimization includes the maximization of Centered Kernel Alignment to highlight a relevant CSP feature set by learning a mapping matrix that matches the best all extracted feature vectors with the provided binary-class label set. As seen in Fig. 1, the proposed κ-FB approach outperforms the classifier accuracy reached by other compared methods in most of the subjects. Moreover, the accuracy averaged across all subjects is as follows: FB - 82.7 ± 5.8, s-FB - 82.7 ± 5.5, and κ-FB 84.9 ± 5.9. Note that the proposal improves the average accuracy without compromising the standard deviation (i.e., confidence) of presented results.

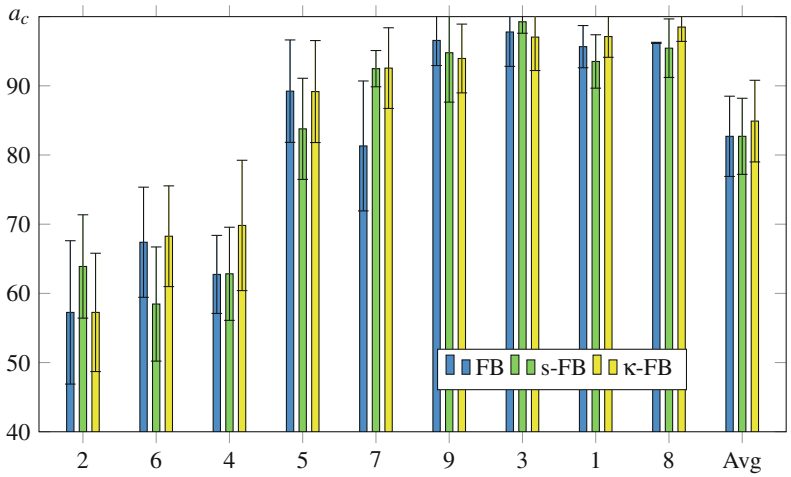


Fig. 1. Testing accuracy results per subject on the test set for all considered CSP-based approaches for motor imagery classification task.

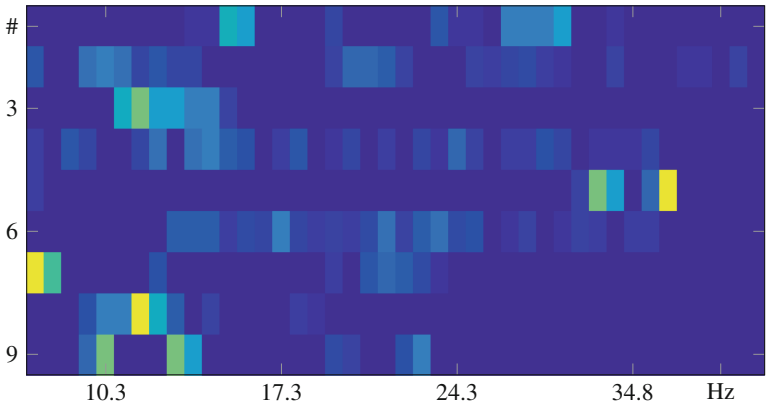
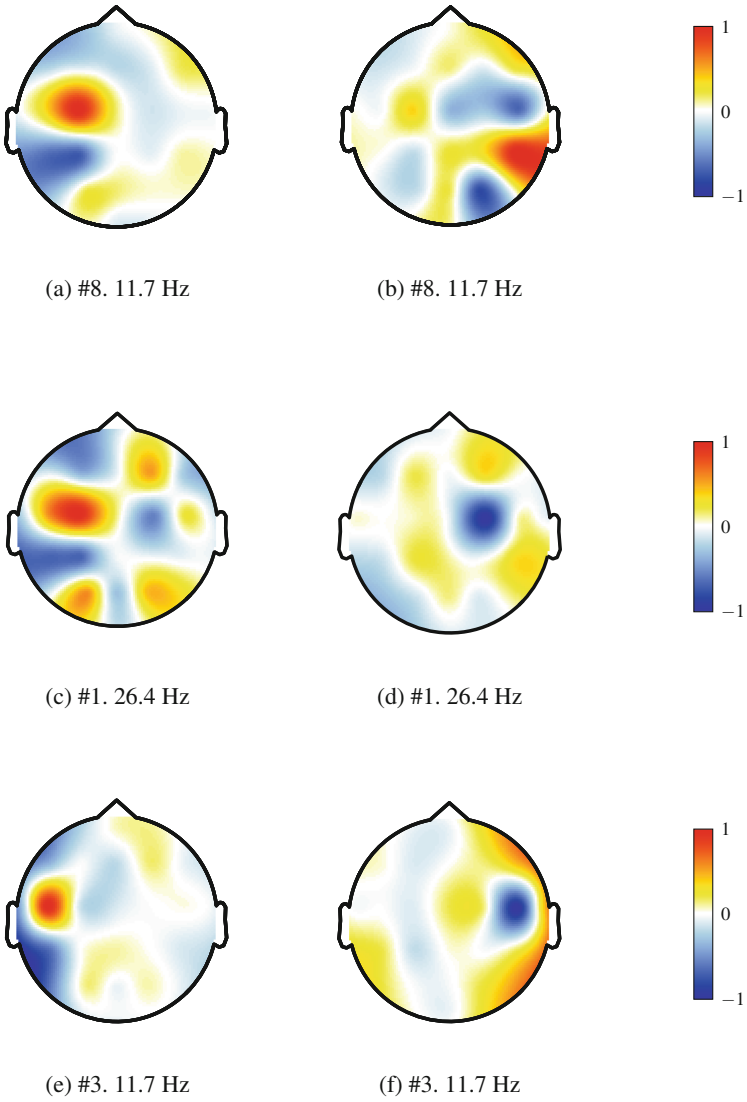


Fig. 2. MKL weights along the frequency bands computed per subject.

The estimated mixing vector allows improving the interpretability, identifying the relevant frequency bands regarding the classifier accuracy. For the sake of illustration, Fig. 2 visually depicts the relative relevance magnitude of each spectral band per subject. Thus, those subjects focalizing the relevance on a less number of frequency bands achieve a better MI discrimination, namely, #8, # 3, #1. At the same time, the subject #5 holds an entirely different spectral behavior, suggesting that he manages the MI task differently. By contrast, the subjects noted as #6, #4, #2 yield a worse accuracy, spreading the relevance contribution on a more considerable amount of frequency bands.



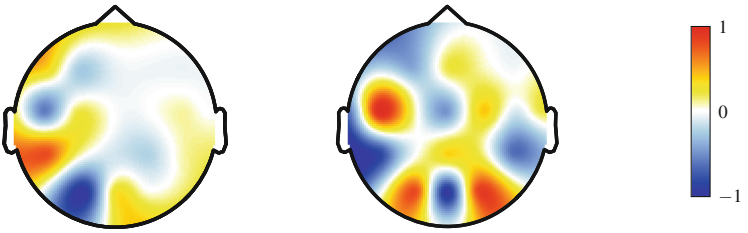
(a) #8. 11.7 Hz (b) #8. 11.7 Hz

(c) #1. 26.4 Hz (d) #1. 26.4 Hz

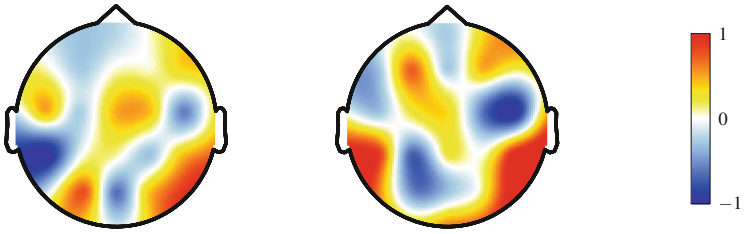
(e) #3. 11.7 Hz (f) #3. 11.7 Hz

Fig. 3. Estimated spatial filters for the best performing subjects (#8, #1, #3) in the most weighted frequency band. Left: first filter ($k = 1$). Right: last filter ($k = 2K$).

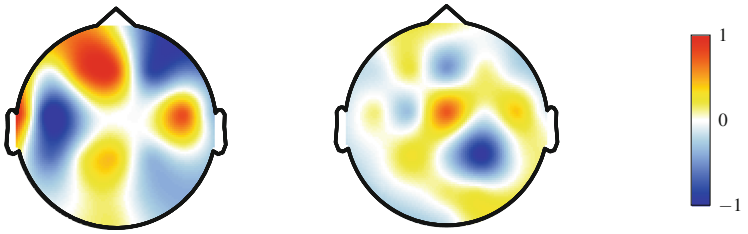
As seen from the topoplots in Fig. 3, the proposed κ-FB method highlights more discriminative information contained in the μ band, more precisely, localized around of the central cortex; this brain region has been associated before with planning and executing of voluntary movements [7]. Besides, the activation surrounding the frontoparietal area ($FC3$, $FC4$, $CP3$, and $CP4$) is related to the subject thinking strategy during MI tasks [6]. On the contrary, the topoplots



(a) #2. 10.3 Hz (b) #2. 10.3 Hz



(c) #6. 23.6 Hz (d) #6. 23.6 Hz



(e) #4. 12.4 Hz (f) #4. 12.4 Hz

Fig. 4. Estimated spatial filters for the worst performing subjects (#2, #6, #4) in the most weighted frequency band. Left: first filter ($k = 1$). Right: last filter ($k = 2K$).

Fig. 4 reveal that the spatial filters associated with the largest-weighted frequency band involve most of the brain areas for the worst performing subjects. This fact is due to the lack of discriminant information in the motor cortex, which may be related to disconnected electrodes, signal artifacts, and subject tiredness or lack of attention [4].

As a concluding remark, the proposed κ-FB reflects the contribution of each frequency band to highlight the spatial filters that better discriminate the MI task, instead of selecting a single band that may not encode all the contribution of the CSP feature set. The use of the kernel alignment to match data and label spaces results in a set of relevance weights that provide enhanced interpretability of performed accuracy by each subject.

For the future work, we plan to extend the kernel learning to other augmented representations, by instance, the one that includes some sliding windowing strategies, aiming to more adequately extract information from time-varying event-related potentials and improving the discrimination of MI tasks. Also, other kernel combination strategies, as tensor-based, can be explored to interpret discriminating information extracted from subjects with a complicated execution of MI tasks. Finally, we plan to improve the parameter tuning stage using methods as Bayesian or heuristic optimization.

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